Отчет по лабораторной работе №4

Архитектура компьютера

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1 Цель работы

Получение навыков правильной работы с репозиториями git.

2 Задание

1. Выполнить работу для тестового репозитория.
2. Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

3 Выполнение лабораторной работы

1. Устанавливаю gitflow. (рис. 3.1).

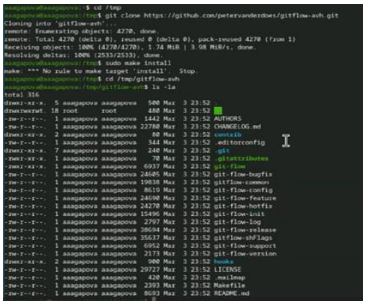

Рис. 3.1: Установка gitflow

2. Устанавливаю gitflow. (рис. 3.2).

```
aaagapova@aaagapova:/tmp/gitflow-avh$ sudo make install
install -d -m 0755 /usr/local/bin
install -d -m 0755 /usr/local/share/doc/gitflow/hooks
install -m 0755 git-flow /usr/local/bin
install -m 0644 git-flow-init git-flow-feature git-flow-bugfix git-flow-hotfix git-flow-release git-flow
-support git-flow-version git-flow-log git-flow-config gitflow-common gitflow-shflags /usr/local/bin
install -m 0644 hooks/filter-flow-hotfix-finish-tag-message hooks/filter-flow-hotfix-start-version hooks
/filter-flow-release-branch-tag-message hooks/filter-flow-release-finish-tag-message hooks/filter-flow-r
elease-start-version hooks/post-flow-bugfix-delete hooks/post-flow-bugfix-finish hooks/post-flow-bugfix-
publish hooks/post-flow-bugfix-pull hooks/post-flow-bugfix-start hooks/post-flow-bugfix-track hooks/post
-flow-feature-delete hooks/post-flow-feature-finish hooks/post-flow-feature-publish hooks/post-flow-feat
ure-pull hooks/post-flow-feature-start hooks/post-flow-feature-track hooks/post-flow-hotfix-delete hooks
/post-flow-hotfix-finish hooks/post-flow-hotfix-publish hooks/post-flow-hotfix-start hooks/post-flow-rel
ease-branch hooks/post-flow-release-delete hooks/post-flow-release-finish hooks/post-flow-release-publis
h hooks/post-flow-release-start hooks/post-flow-release-track hooks/pre-flow-feature-delete hooks/pre-fl
ow-feature-finish hooks/pre-flow-feature-publish hooks/pre-flow-feature-pull hooks/pre-flow-feature-star
t hooks/pre-flow-feature-track hooks/pre-flow-hotfix-delete hooks/pre-flow-hotfix-finish hooks/pre-flow-
hotfix-publish hooks/pre-flow-hotfix-start hooks/pre-flow-release-branch hooks/pre-flow-release-delete h
ooks/pre-flow-release-finish hooks/pre-flow-release-publish hooks/pre-flow-release-start hooks/pre-flow-
release-track /usr/local/share/doc/gitflow/hooks
```

Рис. 3.2: Установка gitflow

3. Проверяю версию gitflow. (рис. 3.3).

```
aaagapova@aaagapova:/tmp/gitflow-avh$ git flow version
1.12.4-dev0 (AVH Edition)
```

Рис. 3.3: Версия gitflow

4. Устанавливаю node.js. (рис. 3.4).

```
[illegible]:/tmp/gitflow-ext$ cd
[illegible]:~$ sudo dnf install nodejs
Updating and loading repositories:
Repositories loaded.
Package                         Arch      Version                        Repository          Size
Installing:
 nodejs                         x86_64    1:22.22.0-2.fc43               updates          161.6 KiB
Installing dependencies:
 nodejs-libs                    x86_64    1:22.22.0-2.fc43               updates           78.1 MiB
Installing weak dependencies:
 nodejs-docs                    noarch    1:22.22.0-2.fc43               updates           97.5 MiB
 nodejs-full-i18n               x86_64    1:22.22.0-2.fc43               updates           30.4 MiB
 nodejs-npm                     x86_64    1:10.9.4-1.22.22.0.2.fc43      updates            9.2 MiB

Transaction Summary:
 Installing:         5 packages

Total size of inbound packages is 40 MiB. Need to download 40 MiB.
After this operation, 215 MiB extra will be used (install 215 MiB, remove 0 B).
Is this ok [y/N]: y
[1/5] nodejs-1:22.22.0-2.fc43.x86_64                          100% |  72.9 KiB/s |  51.4 KiB |  00m01s
[2/5] nodejs-npm-1:10.9.4-1.22.22.0.2.fc43.x86_64             100% |   4.7 MiB/s |   2.3 MiB |  00m01s
[3/5] nodejs-docs-1:22.22.0-2.fc43.noarch                     100% |   4.6 MiB/s |   9.1 MiB |  00m02s
[4/5] nodejs-libs-1:22.22.0-2.fc43.x86_64                     100% |   9.1 MiB/s |  19.6 MiB |  00m02s
[5/5] nodejs-full-i18n-1:22.22.0-2.fc43.x86_64                100% |   7.7 MiB/s |   8.5 MiB |  00m01s
--------------------------------------------------------------------------------------------------
[5/5] Total                                                   100% |  14.2 MiB/s |  39.7 MiB |  00m03s
Running transaction
[1/7] Verify package files                                    100% |  37.0   B/s |   5.0   B |  00m00s
[2/7] Prepare transaction                                     100% |   7.0   B/s |   5.0   B |  00m01s
[3/7] Installing nodejs-libs-1:22.22.0-2.fc43.x86_64          100% | 230.5 MiB/s |  78.1 MiB |  00m00s
[4/7] Installing nodejs-1:22.22.0-2.fc43.x86_64               100% |   4.1 MiB/s | 163.7 KiB |  00m00s
[5/7] Installing nodejs-npm-1:10.9.4-1.22.22.0.2.fc43.x86_64  100% |  20.3 MiB/s |   9.6 MiB |  00m00s
[6/7] Installing nodejs-full-i18n-1:22.22.0-2.fc43.x86_64     100% | 239.5 MiB/s |  30.4 MiB |  00m00s
[7/7] Installing nodejs-docs-1:22.22.0-2.fc43.noarch          100% |  11.8 MiB/s |  97.6 MiB |  00m08s
Complete!
```

Рис. 3.4: Установка node.js

5. Устанавливаю pnpm. (рис. 3.5).

```
[illegible]:~$ sudo dnf install pnpm
Updating and loading repositories:
Repositories loaded.
Package                         Arch      Version                        Repository          Size
Installing:
 pnpm                           noarch    10.27.0-1.fc43                 updates           14.9 MiB

Transaction Summary:
 Installing:         1 package

Total size of inbound packages is 3 MiB. Need to download 3 MiB.
After this operation, 15 MiB extra will be used (install 15 MiB, remove 0 B).
Is this ok [y/N]: y
[1/1] pnpm-0:10.27.0-1.fc43.noarch                            100% |   2.6 MiB/s |   2.7 MiB |  00m01s
--------------------------------------------------------------------------------------------------
[1/1] Total                                                   100% |   1.7 MiB/s |   2.7 MiB |  00m02s
Running transaction
[1/3] Verify package files                                    100% |  47.0   B/s |   1.0   B |  00m00s
[2/3] Prepare transaction                                     100% |   2.0   B/s |   1.0   B |  00m00s
[3/3] Installing pnpm-0:10.27.0-1.fc43.noarch                 100% |  13.5 MiB/s |  15.1 MiB |  00m01s
Complete!
```

Рис. 3.5: Установка pnpm

6. Добавляю каталог с исполняемыми файлами в переменную PATH. (рис. 3.6).

```
aaagapova@aaagapova:~$ pnpm setup
Appended new lines to /home/aaagapova/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/aaagapova/.local/share/pnpm"
case ":$PATH:" in
  *":$PNPM_HOME:"*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/aaagapova/.bashrc
```

Рис. 3.6: Добавление каталога

7. Устанавливаю программу для форматирования коммитов. (рис. 3.7).

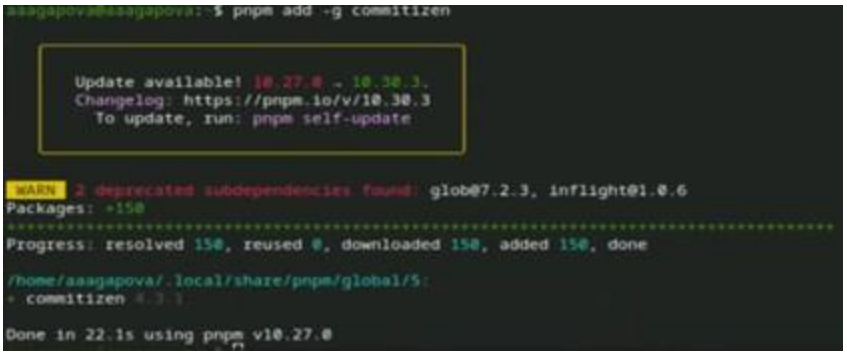

Рис. 3.7: Установка программы

8. Устанавливаю программу для помощи в создании логов. (рис. 3.8).

Рис. 3.8: Установка программы

9. Создаю новый репозиторий. (рис. 3.9).

Рис. 3.9: Создание нового репозитория

10. Создаю папку и инициализирую. (рис. 3.10).

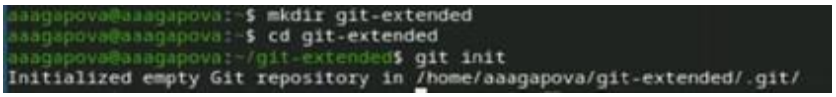

Рис. 3.10: Создание папки

11. Создаю первый файл и делаю коммит. (рис. 3.11).

```
aaagapova@aaagapova:~/git-extended$ touch README.md
aaagapova@aaagapova:~/git-extended$ git add README.md
aaagapova@aaagapova:~/git-extended$ git commit -m "first commit"
[master (root-commit) 1ee9dbf] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 README.md
```

Рис. 3.11: Создание файла и коммита

12. Переименовываю ветку в main, подключаю удаленные репозитории и отправляю на GitHub. (рис. 3.12).

```
aaagapova@aaagapova:~/git-extended$ git branch -M main
aaagapova@aaagapova:~/git-extended$ git remote add origin https://github.com/anyaagapo
vaa/git-extended.git
aaagapova@aaagapova:~/git-extended$ push -u origin main
bash: push: command not found
aaagapova@aaagapova:~/git-extended$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 865 bytes | 216.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anyaagapovaa/git-extended.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Рис. 3.12: Отправление на GitHub

13. Редактирую файл package.json. (рис. 3.13).

```
aaagapova@aaagapova:~/git-extended$ cat package.json
{
    "name": "git-extended",
    "version": "1.0.0",
    "description": "Git repo for educational purposes",
    "main": "index.js",
    "repository": "git@github.com:anyaagapova/git-extended.git",
    "author": "Anna Agapova <aagapova2017@gmail.com>",
    "license": "CC-BY-4.0",
    "config": {
        "commitizen": {
            "path": "cz-conventional-changelog"
        }
    }
}
```

Рис. 3.13: Редактирование файла

14. Добавляю новые файлы, выполняю коммит и отправляю на GitHub. (рис. 3.14).

```
aaagapova@aaagapova:~/git-extended$ git add .
aaagapova@aaagapova:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: feat:     A new feature
? What is the scope of this change (e.g. component or file name): (press enter to
skip)
? Write a short, imperative tense description of the change (max 94 chars):
 (6) update
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[main fbe0a57] feat: update
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
aaagapova@aaagapova:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.14 KiB | 1.14 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anyaagapovaa/git-extended.git
   1ee9dbf..fbe0a57  main -> main
```

Рис. 3.14: Отправление на GitHub

15. Инициализирую gitflow. (рис. 3.15).

```
aaagapova@aaagapova:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
   - main
Branch name for production releases: [main] main
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/aaagapova/git-extended/.git/hooks]
```

Рис. 3.15: Проверка работы

16. Проверяю, что я на ветке develop. (рис. 3.16).

```
aaagapova@aaagapova:~/git-extended$ git branch
* develop
  main
```

Рис. 3.16: Проверка ветки

17. Загружаю весь репозиторий в хранилище. (рис. 3.17).

```
aaagapova@aaagapova:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/anyaagapovaa/git-extended/pull/new/develop
remote:
To https://github.com/anyaagapovaa/git-extended.git
 * [new branch]      develop -> develop
```

Рис. 3.17: Загрузка репозитория

18. Устанавливаю внешнюю ветку как вышестоящую для этой ветки. (рис. 3.18).

Рис. 3.18: Установка ветки

19. Создаю релиз с версией 1.0.0. (рис. 3.19).

```
aaagapova@aaagapova:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

     git flow release finish '1.0.0'
```

Рис. 3.19: Создание релиза

20. Создаю журнал изменений и добавляю журнал изменений в индекс. (рис. 3.20).

```
aaagapova@aaagapova:~/git-extended$ standard-changelog --first-release
aaagapova@aaagapova:~/git-extended$ git add CHANGELOG.md
aaagapova@aaagapova:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 e32faf2] chore(site): add changelog
 1 file changed, 6 insertions(+)
 create mode 100644 CHANGELOG.md
```

Рис. 3.20: Создание журнала

21. Заливаю релизную ветку в основную ветку. (рис. 3.21).

```
aaagapova@aaagapova:~/git-extended$ git flow release finish 1.0.0
Branches 'main' and 'origin/main' have diverged.
And local branch 'main' is ahead of 'origin/main'.
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 6 ++++++
 1 file changed, 6 insertions(+)
 create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was e32faf2).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'main'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'
```

Рис. 3.21: Завершение релиза

22. Отправляю данные на GitHub. (рис. 3.22).

```
aaagapova@aaagapova:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.78 KiB | 2.78 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anyaagapovaa/git-extended.git
   fbe0a57..d97f947  develop -> develop
   fbe0a57..c9746ed  main -> main
aaagapova@aaagapova:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 164 bytes | 164.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anyaagapovaa/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
```

Рис. 3.22: Отправление на GitHub

23. Создаю релиз на github. Для этого буду использовать утилиты работы с github. (рис. 3.23).

```
aaagapova@aaagapova:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/anyaagapovaa/git-extended/releases/tag/v1.0.0
```

Рис. 3.23: Создание релиза

24. Создаю ветку для новой функциональности. (рис. 3.24).

```
aaagapova@aaagapova:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

     git flow feature finish feature_branch
```

Рис. 3.24: Создание ветки

25. Объединяю ветку feature_branch с develop. (рис. 3.25).

```
aaagapova@aaagapova:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was d97f947).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'
```

Рис. 3.25: Объединение веток

26. Создаю релиз с версией 1.2.3. (рис. 3.26).

```
aaagapova@aaagapova:~/git-extended$ git flow release start 1.2.3
Switched to a new branch 'release/1.2.3'

Summary of actions:
- A new branch 'release/1.2.3' was created, based on 'develop'
- You are now on branch 'release/1.2.3'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

     git flow release finish '1.2.3'
```

Рис. 3.26: Создание релиза

27. Обновляю номер версии в файле package.json. (рис. 3.27).

```
{
    "name": "git-extended",
    "version": "1.2.3",
    "description": "Git repo for educational purposes",
    "main": "index.js",
    "repository": "git@github.com:anyaagapova/git-extended.git",
    "author": "Anna Agapova <aagapova2017@gmail.com>",
    "license": "CC-BY-4.0",
    "config": {
        "commitizen": {
            "path": "cz-conventional-changelog"
        }
    }
}
```

Рис. 3.27: Обновление файла

28. Создаю журнал изменений и добавляю журнал изменений в индекс (рис. 3.28).

```
aaagapova@aaagapova:~/git-extended$ standard-changelog
aaagapova@aaagapova:~/git-extended$ git add CHANGELOG.md
aaagapova@aaagapova:~/git-extended$ git commit -am 'chore(site): update changelog'
[release/1.2.3 dfdea8b] chore(site): update changelog
 2 files changed, 3 insertions(+), 1 deletion(-)
```

Рис. 3.28: Создание журнала

29. Заливаю релизную ветку в основную ветку. (рис. 3.29).

```
aaagapova@aaagapova:~/git-extended$ git flow release finish 1.2.3
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 2 ++
 package.json | 2 +-
 2 files changed, 3 insertions(+), 1 deletion(-)
Already on 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 2 ++
 package.json | 2 +-
 2 files changed, 3 insertions(+), 1 deletion(-)
Deleted branch release/1.2.3 (was dfdea8b).

Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'main'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'
```

Рис. 3.29: Завершение релиза

30. Отправляю данные на GitHub. (рис. 3.30).

```
aaagapova@aaagapova:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.77 KiB | 708.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/anyaagapovaa/git-extended.git
   d97f947..5e449cc  develop -> develop
   c9746ed..d3a4b58  main -> main
aaagapova@aaagapova:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 159 bytes | 159.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anyaagapovaa/git-extended.git
 * [new tag]         v1.2.3 -> v1.2.3
```

Рис. 3.30: Отправление на GitHub

31. Создаю релиз на github с комментарием из журнала изменений. (рис. 3.31).

```
aaagapova@aaagapova:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/anyaagapovaa/git-extended/releases/tag/v1.2.3
```

Рис. 3.31: Создание релиза

32. Проверяю, что все файлы создались. (рис. 3.32).

Рис. 3.32: Проверка

33. Проверяю, что релизы создались. (рис. 3.33).

Рис. 3.33: Проверка

4 Выводы

Я получила навыки правильной работы с репозиториями git.

Список литературы